Lecture Unit 3.1

Introduction to PDEs

What is a differential equation?

- Many HPC applications require calculating numerical solutions to differential equations
- In unit 1.1 we saw examples from weather and climate, astrophysics, and engineering
- A differential equation (DE) is an equation which contains one or more derivatives
- Derivatives describe the rate of change of one variable with respect to another
- For example velocity is the rate of change of position with respect to time: $v = \frac{dx}{dt}$

What is a differential equation?

- A partial differential equation (PDE) is a differential equation which contains derivatives with respect to more than one variable

- For example if a variable u depends on two variables x and t we write $u = u(x,t)$

  The rate of change of u with respect to t is given by $\dfrac{\partial u}{\partial t}$

  The rate of change of u with respect to x is given by $\dfrac{\partial u}{\partial x}$

- We write the derivatives with curly ∂'s to show that we are considering the rate of change with respect to one variable whilst keeping the other variable(s) constant

Time derivative

- We will look at DEs with a time derivative on the left hand side
- The rate of change of a variable u with respect to time is given by $\dfrac{du}{dt}$
- Examples:
  - Speed or velocity is the rate of change of position (e.g. kilometres per hour, m/s)
  - Acceleration is the rate of change of velocity (e.g. m/s^2)
  - Cooling rate of a hot object (e.g. °C/s)

Spatial derivatives

- PDEs often involve spatial gradients (rate of change of some property in space) as well as a time derivative

- For example the motion of a fluid depends on pressure gradients

- Spatial derivative describe the rate of change of a variable with respect to a spatial co-ordinate (position)

- The rate of change of a variable u with respect to position in the x direction is given by $\dfrac{\partial u}{\partial x}$

Unit Summary

- Many HPC applications involve calculating numerical solutions to partial differential equations
- A differential equation contains one or more derivatives (rates of change)
- Partial differential equations contain derivatives with respect to more than one variable
- In this module we will be considering PDEs with time and space derivatives